

STA5228 Reliability and Life Data Project

Reliability Assessment of Dishwashers. A Multimodal Failure Analysis

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- When assessing the reliability or survival of a product very often we have to deal with more than one failing mode.
- Each failure mode is like a component in a series system. When one of the components fail the system fails.
- This project's concern is the reliability of dishwashers.
- We are going to do a parametric and non-parametric analysis on the reliability of the dishwashers of our data set.

- Perform a parametric analysis taking into consideration each failure mode separately while considering the other failure mode times as right censored times.
- We do a non-parametric analysis, giving Kaplan-Meier estimates of the survival probabilities for both cases aforementioned.
- We estimate the reliability taking in consideration both failure modes.

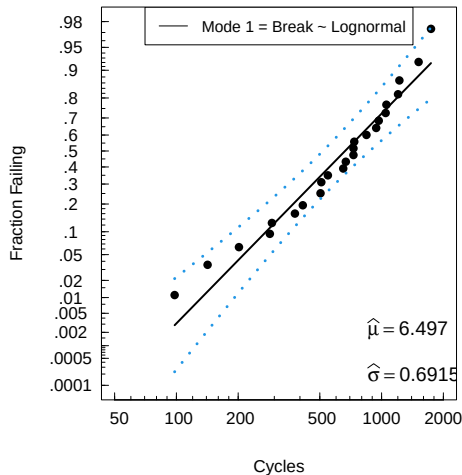
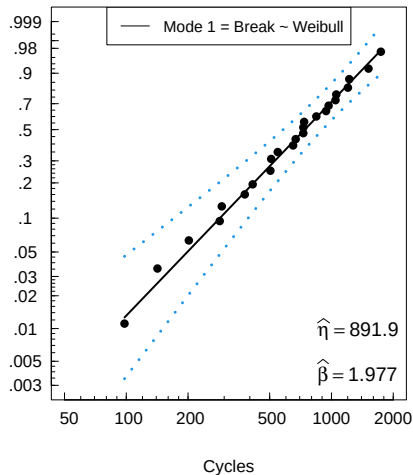
Parametric Analysis of the Dishwasher data

- We have two failure modes, break and obstruction.
- We assume both failure modes are independent from each other.
- The failures for failure mode j , are marked as failures and the failures for the other failure mode are marked as right-censored observations.

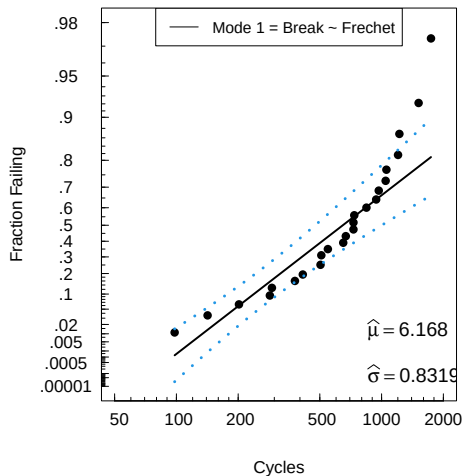
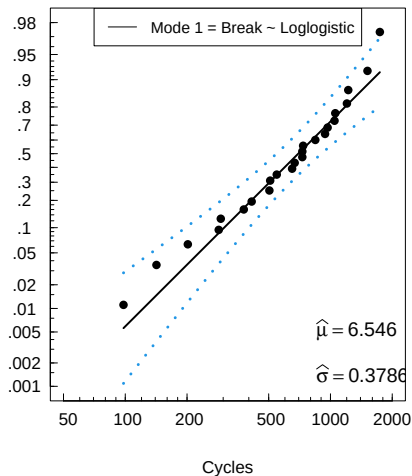
2.1 Parametric Analysis for Failure Mode 1 = Break

- First we consider the failure mode break.
- Failures for failure mode break are considered as failures and the failures for failure mode obstruction are marked as right-censored observations.
- We need to find a distribution that fits our data. For this, we use probability plots.

2.1 Parametric Analysis for Failure Mode 1 = Break



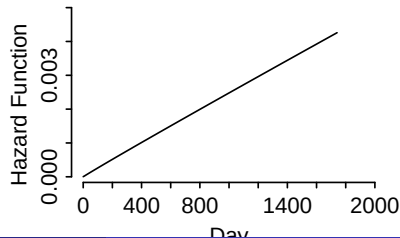
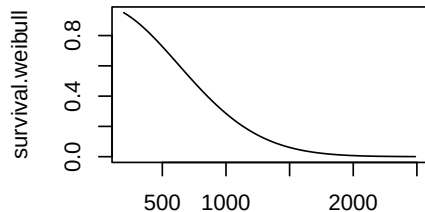
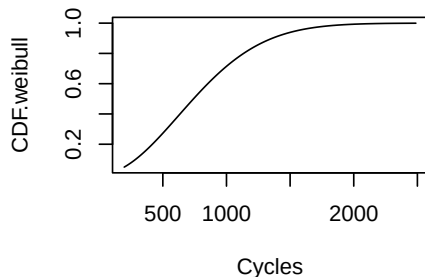
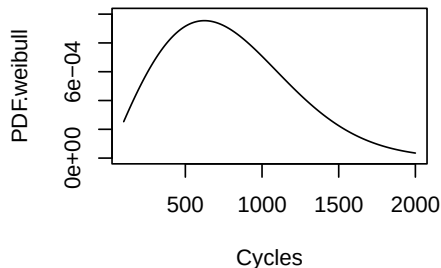
2.1 Parametric Analysis for Failure Mode 1 = Break



2.1 Parametric Analysis for Failure Mode 1 = Break

- From these plots we see that for this failure mode Weibull distribution is a better fit.
- Thus we are going to use this distribution for the parametric analysis.
- In the next figure we have the pdf, CDF, survival and hazard function for the Weibull distribution with parameters, the ML estimates.

2.1 Parametric Analysis for Failure Mode 1 = Break



2.1 Parametric Analysis for Failure Mode 1 = Break

- Under the assumption that the Weibull distribution is correct, we have
- The ML estimate for the Mean Time To Failure is $MTTF = 790.6$ cycles
- We can see that 10% of the dishwashers will fail due to arm spray break after about 87 cycles
- and 90% of them will fail after about 1360 cycles

2.1 Parametric Analysis for Failure Mode 1 = Break

- After 1000 cycles the probability that the spray arm will break is 0.7145, with a 95% confidence interval (0.5712, 0.8437).
- The hazard rate is very small, which tells us that in general the spray arms take longer to break, after having completed many cycles.
- ML Estimates table

##	MLE	Std.Err.	95% Lower	95% Upper
## mu	6.7934	0.1018	6.5938	6.9930
## sigma	0.5059	0.0708	0.3845	0.6655
## Weibull (eta)	891.9292	90.8269	730.5514	1088.9550
## Weibull (beta)	1.9767	0.2766	1.5026	2.6004

2.1 Parametric Analysis for Failure Mode 1 = Break

##	Cycles	Fhat	Std.Err.	95% Lower	95% Upper
## [1,]	50	0.003355	0.002826	0.0006428	0.01741
## [2,]	100	0.013140	0.008603	0.0036280	0.04700
## [3,]	200	0.050730	0.023720	0.0201200	0.12490
## [4,]	500	0.272800	0.062770	0.1708000	0.41830
## [5,]	1000	0.714500	0.071640	0.5712000	0.84370
## [6,]	2000	0.992800	0.010060	0.9411000	0.99980

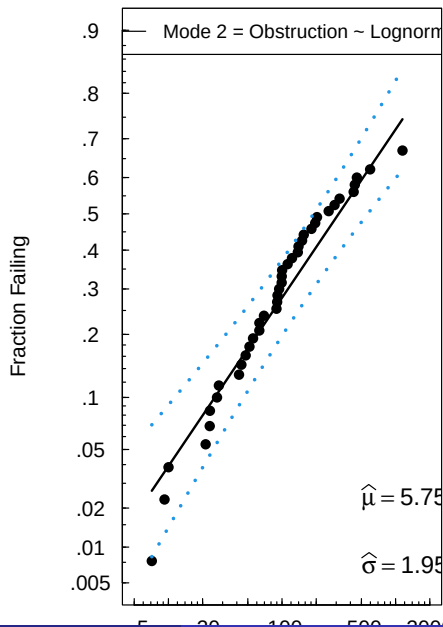
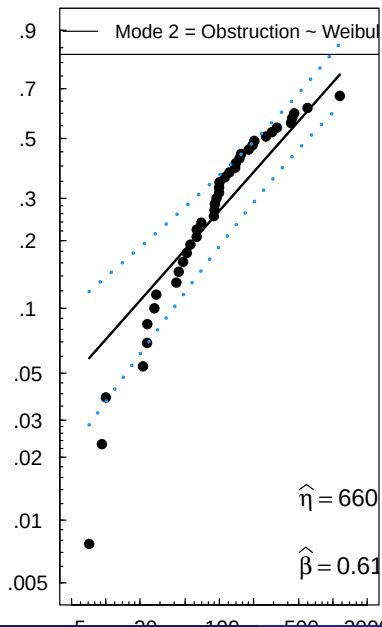
2.1 Parametric Analysis for Failure Mode 1 = Break

##		p	Quanhat	Std.Err.	95% Lower	95% Upper
##	[1,]	0.001	27.09	13.83	9.958	73.69
##	[2,]	0.005	61.21	24.45	27.980	133.90
##	[3,]	0.010	87.03	30.63	43.650	173.50
##	[4,]	0.050	198.50	48.39	123.100	320.10
##	[5,]	0.100	285.70	56.72	193.600	421.60
##	[6,]	0.200	417.60	64.82	308.100	566.10
##	[7,]	0.300	529.50	69.79	408.900	685.50
##	[8,]	0.400	635.00	74.39	504.700	798.90
##	[9,]	0.500	741.00	79.95	599.700	915.50
##	[10,]	0.600	853.30	87.65	697.700	1044.00
##	[11,]	0.700	979.70	99.14	803.500	1195.00
##	[12,]	0.800	1135.00	117.50	926.200	1390.00
##	[13,]	0.900	1360.00	152.00	1093.000	1693.00
##	[14,]	0.990	1931.00	270.10	1468.000	2540.00

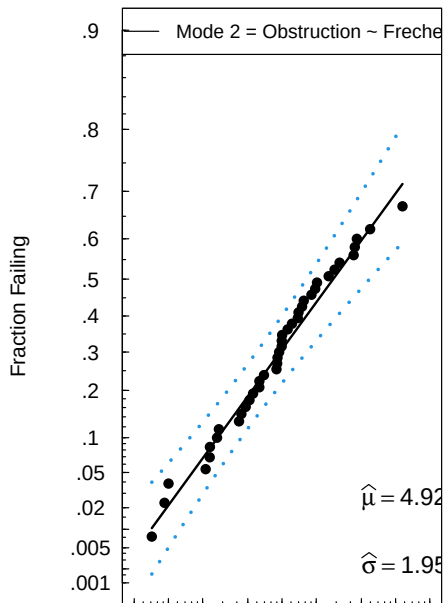
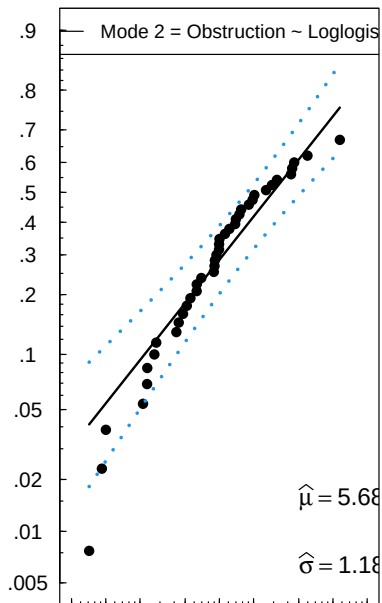
2.2 Parametric Analysis for Failure Mode 2 = Obstruction

- Now we consider the failure mode obstruction.
- Failures for failure mode obstruction as failures and the failures for failure mode break are marked as right-censored observations.
- For the parametric analysis we need to find a distribution that fits well our data.
- Next slide we have the probability plots for four different distributions.

2.2 Parametric Analysis for Failure Mode 2 = Obstruction



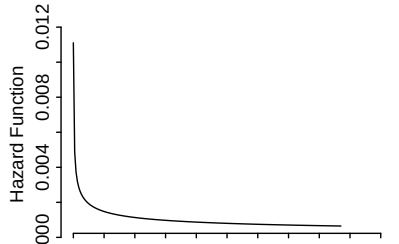
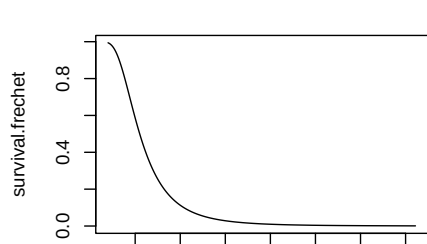
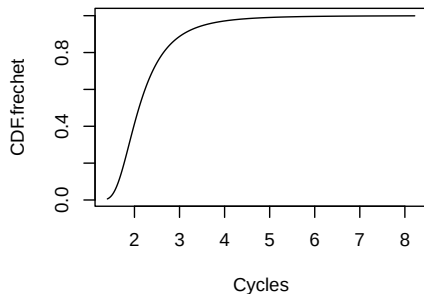
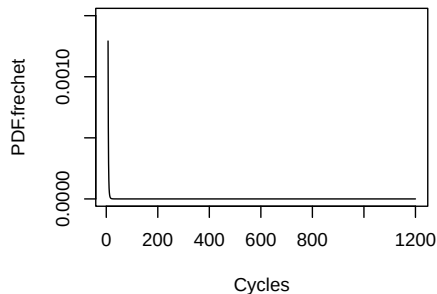
2.2 Parametric Analysis for Failure Mode 2 = Obstruction



2.2 Parametric Analysis for Failure Mode 2 = Obstruction

- From these four plots, Frechet distributions seems to be a better fit for the failure mode obstruction.
- In the next slide we have the pdf, CDF, survival and hazard function for the Frechet distribution with parameters, the ML estimates.

2.2 Parametric Analysis for Failure Mode 2 = Obstruction



2.2 Parametric Analysis for Failure Mode 2 = Obstruction

- The ML estimate for the Mean Time To Failure under this distribution does not exist, since the shape parameter $\beta = \frac{1}{\sigma} = 0.513 < 1$.
- 10% of the dishwashers will fail due to arm spray obstruction after about 7 cycles
- 90% of them will fail after about 11120 cycles.
- After 1000 cycles the probability that the spray arm will break is 0.6961, with a 95% confidence interval (0.5733, 0.7898).
- The hazard rate is decreasing, with the increase in the number of cycles.

2.2 Parametric Analysis for Failure Mode 2 = Obstruction

##		MLE	Std.Err.	95% Lower	95% Upper
## mu		4.9276	0.2675	4.4034	5.4518
## sigma		1.9504	0.2356	1.5392	2.4714

2.2 Parametric Analysis for Failure Mode 2 = Obstruction

##		Cycles	Fhat	Std.Err.	95% Lower	95% Upper
##	[1,]	5	0.004165	0.004423	0.0003314	0.02353
##	[2,]	10	0.021460	0.013450	0.0050430	0.06143
##	[3,]	20	0.067710	0.025410	0.0290600	0.12890
##	[4,]	50	0.185800	0.039010	0.1165000	0.26770
##	[5,]	100	0.307400	0.047170	0.2182000	0.40080
##	[6,]	200	0.437400	0.053600	0.3310000	0.53880
##	[7,]	500	0.596400	0.057070	0.4757000	0.69800
##	[8,]	1000	0.696100	0.055180	0.5733000	0.78980
##	[9,]	2000	0.775700	0.050280	0.6578000	0.85730

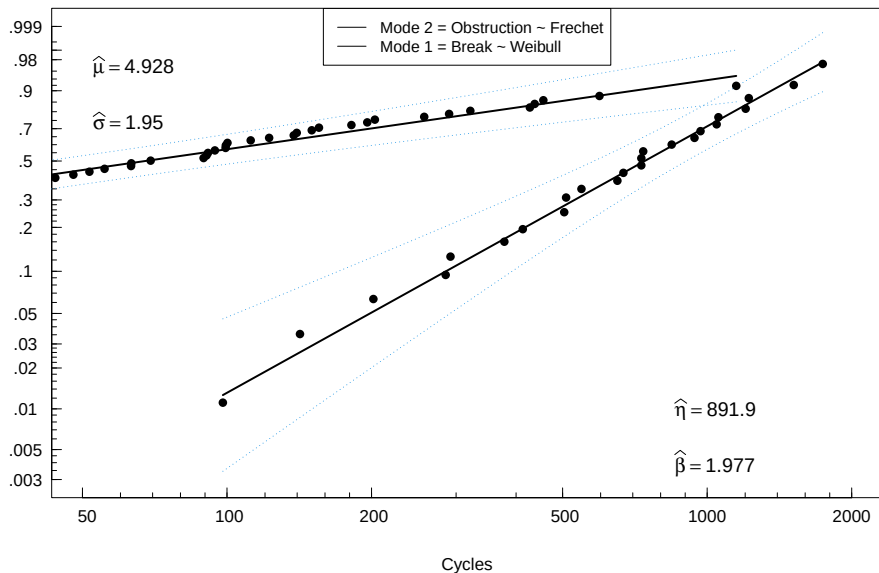
2.2 Parametric Analysis for Failure Mode 2 = Obstruction

##		p	Quanhat	Std.Err.	95% Lower	95% Upper
##	[1,]	0.001	3.184e+00	1.341e+00	1.395	7.267e+00
##	[2,]	0.005	5.342e+00	1.986e+00	2.578	1.107e+01
##	[3,]	0.010	7.022e+00	2.440e+00	3.554	1.387e+01
##	[4,]	0.050	1.624e+01	4.614e+00	9.309	2.834e+01
##	[5,]	0.100	2.714e+01	6.988e+00	16.380	4.495e+01
##	[6,]	0.200	5.457e+01	1.326e+01	33.890	8.787e+01
##	[7,]	0.300	9.611e+01	2.426e+01	58.610	1.576e+02
##	[8,]	0.400	1.637e+02	4.530e+01	95.180	2.816e+02
##	[9,]	0.500	2.821e+02	8.844e+01	152.600	5.215e+02
##	[10,]	0.600	5.117e+02	1.858e+02	251.100	1.043e+03
##	[11,]	0.700	1.031e+03	4.431e+02	444.000	2.394e+03
##	[12,]	0.800	2.574e+03	1.350e+03	920.400	7.196e+03
##	[13,]	0.900	1.112e+04	7.632e+03	2898.000	4.268e+04
##	[14,]	0.990	1.088e+06	1.327e+06	99600.000	1.188e+07

2.2 Parametric Analysis for Failure Mode 2 = Obstruction

- We construct a multiple probability plot that shows the parametric and non-parametric estimates for each of the failure modes together in one plot.

2.2 Parametric Analysis for Failure Mode 2 = Obstruction



3. Non-Parametric Analysis

- Kaplan-Meier Estimators for both failure modes

3.1 Non-Parametric Analysis for Failure Mode 1 = Break

- We are going to compute the Kaplan-Meier estimates for the failing and survival probabilities.
- The table below gives the Kaplan-Meier Estimates

3.1 Non-Parametric Analysis for Failure Mode 1 = Break

- The top part of the table for the Kaplan-Meier estimates

##	Failure	Cycles	S.hat	F.hat
## 21	Break	98.04	0.9777778	0.02222222
## 29	Break	141.90	0.9513514	0.04864865
## 34	Break	201.78	0.9216216	0.07837838
## 37	Break	285.38	0.8898416	0.11015843
## 39	Break	292.05	0.8568845	0.14311553
## 41	Break	378.10	0.8226091	0.17739091

3.1 Non-Parametric Analysis for Failure Mode 1 = Break

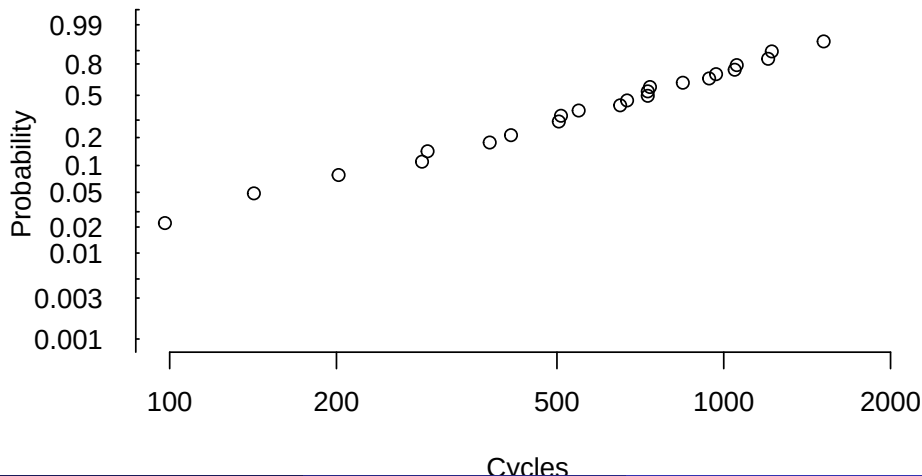
- Here we have the bottom part of the table

##	Failure	Cycles	S.hat	F.hat
## 58	Break	1046.52	0.25226679	0.7477332
## 59	Break	1055.45	0.21022232	0.7897777
## 61	Break	1202.70	0.15766674	0.8423333
## 62	Break	1221.00	0.10511116	0.8948888
## 63	Break	1514.70	0.05255558	0.9474444
## 64	Break	1740.75	0.00000000	1.0000000

3.1 Non-Parametric Analysis for Failure Mode 1 = Break

- We construct a probability plot for the failures of failure mode break.

Weibull Probability Scale



3.1 Non-Parametric Analysis for Failure Mode 1 = Break

- Here we use the survival function to estimate the survival probabilities for each cycle and the corresponding 95% confidence intervals.

##	Cycles	Survival	Std.err	Lower 95%	Upper 95%
## [1,]	98.04	0.9772727	0.02246752	0.9342146	1.0000000
## [2,]	141.90	0.9501263	0.03454845	0.8847691	1.0000000
## [3,]	201.78	0.9194770	0.04502116	0.8353390	1.0000000
## [4,]	285.38	0.8866386	0.05407923	0.7867356	0.9992276
## [5,]	292.05	0.8525371	0.06182319	0.7395830	0.9827422
## [6,]	378.10	0.8170147	0.06869859	0.6928779	0.9633920

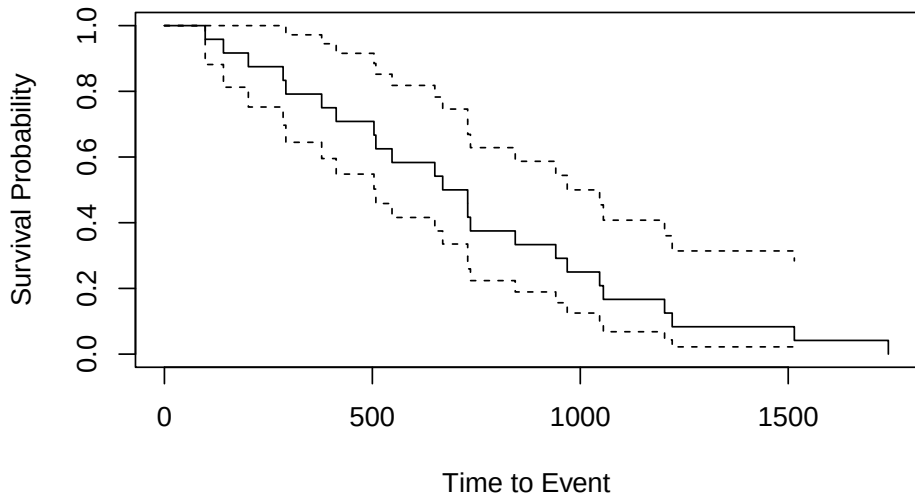
3.1 Non-Parametric Analysis for Failure Mode 1 = Break

##		Cycles	Survival	Std.err	Lower 95%	Upper 95%
##	[19,]	1046.52	0.26323952	0.09077565	0.133912871	0.5174637
##	[20,]	1055.45	0.21936627	0.08559454	0.102103219	0.4713031
##	[21,]	1202.70	0.16452470	0.07985495	0.063545604	0.4259677
##	[22,]	1221.00	0.10968313	0.06956439	0.031643308	0.3801875
##	[23,]	1514.70	0.05484157	0.05209222	0.008522778	0.3528893
##	[24,]	1740.75	0.00000000	NaN	NA	NA

- The two different ways of computing the survival probabilities give similar results

3.1 Non-Parametric Analysis for Failure Mode 1 = Break

Kaplan–Meier Survival Curve



3.2 Non-Parametric Analysis for Failure Mode 2 = Obstruction

- We repeat the same process as for failure mode 1 = break.
- Compute the Kaplan-Meier estimators, and construct a probability plot for obstruction failure times.

3.2 Non-Parametric Analysis for Failure Mode 2 = Obstruction

- Top part of the table for the Kaplan-Meier estimates

##	Failure	Cycles	S.hat1	F.hat1
## 1	Obstruction	7.14	0.9846154	0.01538462
## 2	Obstruction	9.24	0.9694675	0.03053254
## 3	Obstruction	10.02	0.9540791	0.04592092
## 4	Obstruction	21.27	0.9386907	0.06130929
## 5	Obstruction	23.10	0.9233023	0.07669766
## 6	Obstruction	23.19	0.9079140	0.09208603

3.2 Non-Parametric Analysis for Failure Mode 2 = Obstruction

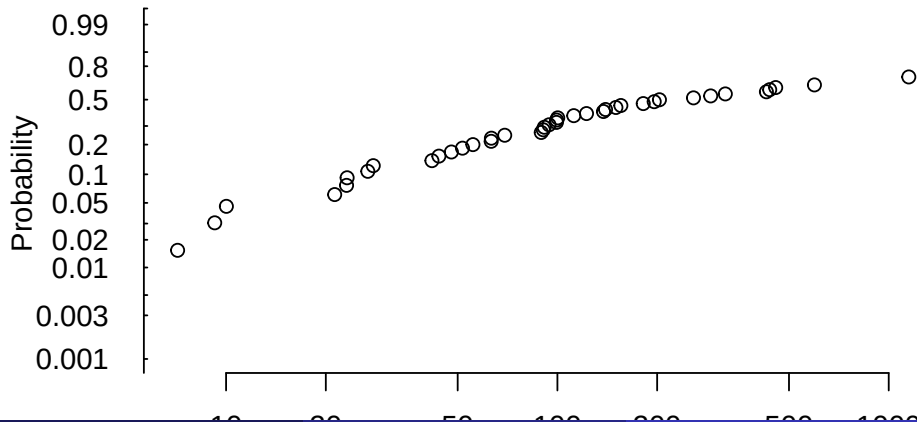
- Bottom part of the table

##	Failure	Cycles	S.hat1	F.hat1
## 1	Obstruction	7.14	0.9846154	0.01538462
## 2	Obstruction	9.24	0.9694675	0.03053254
## 3	Obstruction	10.02	0.9540791	0.04592092
## 4	Obstruction	21.27	0.9386907	0.06130929
## 5	Obstruction	23.10	0.9233023	0.07669766
## 6	Obstruction	23.19	0.9079140	0.09208603

3.2 Non-Parametric Analysis for Failure Mode 2 = Obstruction

- Probability plot for the failures of failure mode obstruction.

Weibull Probability Scale



3.2 Non-Parametric Analysis for Failure Mode 2 = Obstruction

- Using the survival function to estimate the survival probabilities for each cycle and the corresponding 95% confidence intervals.

##		Cycles	Survival	Std.err	Lower 95%	Upper 95%
##	[1,]	7.14	0.984375	0.01550245	0.9544549	1.0000000
##	[2,]	9.24	0.968750	0.02174908	0.9270468	1.0000000
##	[3,]	10.02	0.953125	0.02642139	0.9027217	1.0000000
##	[4,]	21.27	0.937500	0.03025768	0.8800328	0.9987199
##	[5,]	23.10	0.921875	0.03354602	0.8584159	0.9900254
##	[6,]	23.19	0.906250	0.03643507	0.8375797	0.9805504

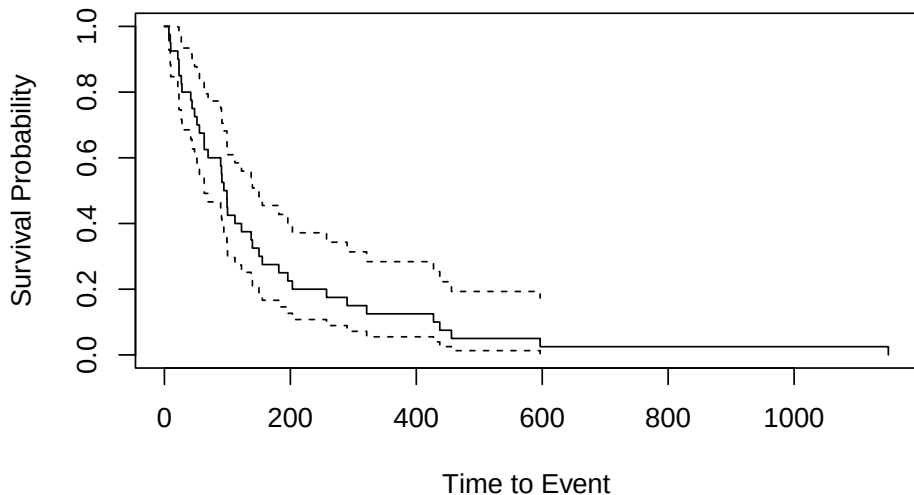
3.2 Non-Parametric Analysis for Failure Mode 2 = Obstruction

##		Cycles	Survival	Std.err	Lower 95%	Upper 95%
##	[35,]	321.20	0.4398636	0.06341209	0.3315936	0.5834852
##	[36,]	427.35	0.4198698	0.06360368	0.3120117	0.5650128
##	[37,]	437.29	0.3998760	0.06363992	0.2927233	0.5462523
##	[38,]	455.87	0.3798822	0.06352105	0.2737268	0.5272061
##	[39,]	596.67	0.3561395	0.06383417	0.2506406	0.5060447
##	[40,]	1149.66	0.2849116	0.08164928	0.1624707	0.4996265

- The two different ways of computing the survival probabilities give similar results

3.2 Non-Parametric Analysis for Failure Mode 2 = Obstruction

Kaplan–Meier Survival Curve



4. Consider both Failure Modes

- Let T_1 and T_2 be the failure times for failure mode 1 and failure mode 2 respectively.
- Then, the joint distribution of T_1 and T_2 is $F(t_1, t_2) = \Pr(T_1 \leq t_1, T_2 \leq t_2)$ and it completely describes the multiple failure modes probability model.
- The marginal distribution for mode j is $F_j(t) = \Pr(T_j \leq t)$.
- We can interpret this as the cdf for mode j if only mode j is active, where $j = 1, 2$ in our case.
- We assume that failure modes are independent from each other.

4. Consider both Failure Modes

- The joint distribution of the two failure modes is

$$F(t_1, t_2) = \prod_{j=1}^J F_j(t_j)$$

- The system failure time is the first failure time.

-

$$T = \min(T_1, T_2)$$

-

$$F_T(t) = \Pr(T \leq t) = 1 - \Pr(T > t) = 1 - \Pr(T_1 > t_1 \cap T_2 > t_2)$$

$$F_T(t) = 1 - \prod_{j=1}^2 [1 - F_j(t)]$$

4. Consider both Failure Mode

- For failure mode 1 = break we assumed that the failures followed a Weibull distribution
- For failure mode 2 = obstruction, we assumed a Frechet distribution.
- The system's failure-time cdf

-

$$F_T(t) = 1 - \exp \left[- \left(\frac{t}{\eta_1} \right)^{\beta_1} \right] + \exp \left[- \left(\frac{t}{\eta_1} \right)^{\beta_1} - \left(\frac{t}{\eta_2} \right)^{-\beta_2} \right]$$

- where, $\eta_2 = \exp(\mu)$ and $\beta_2 = \frac{1}{\sigma}$.

4. Consider both Failure Mode

- An estimate of the failure-time cdf for the system with all failure modes active is obtained by substituting the ML estimates for the parameters for all of the failure modes into the system cdf.
- Since we already obtained the ML estimates for all the failure modes, we can estimate the cdf of the system failure-time.
- The ML estimates are :

4. Consider both Failure Mode

##		Cycles	F.hat.joint	S.hat.joint
##	[1,]	7.14	0.01044737	0.9895526
##	[2,]	9.24	0.01838823	0.9816118
##	[3,]	10.02	0.02164128	0.9783587
##	[4,]	21.27	0.07411156	0.9258884
##	[5,]	23.10	0.08260727	0.9173927
##	[6,]	23.19	0.08302153	0.9169785

4. Consider both Failure Mode

##		Cycles	F.hat.joint	S.hat.joint
##	[59,]	1055.45	0.9263869	0.073613068
##	[60,]	1149.66	0.9451166	0.054883366
##	[61,]	1202.70	0.9538608	0.046139241
##	[62,]	1221.00	0.9566029	0.043397122
##	[63,]	1514.70	0.9853055	0.014694490
##	[64,]	1740.75	0.9943938	0.005606226

4. Consider both Failure Mode

- Kaplan-Meier estimates for system failure-times

4. Consider both Failure Mode

- Top part of the table for the Kaplan-Meier estimates

##	Cycles	F.hat.KM	S.hat.KM
## [1,]	7.14	0.01538462	0.9846154
## [2,]	9.24	0.03076923	0.9692308
## [3,]	10.02	0.04615385	0.9538462
## [4,]	21.27	0.06153846	0.9384615
## [5,]	23.10	0.07692308	0.9230769
## [6,]	23.19	0.09230769	0.9076923

4. Consider both Failure Mode

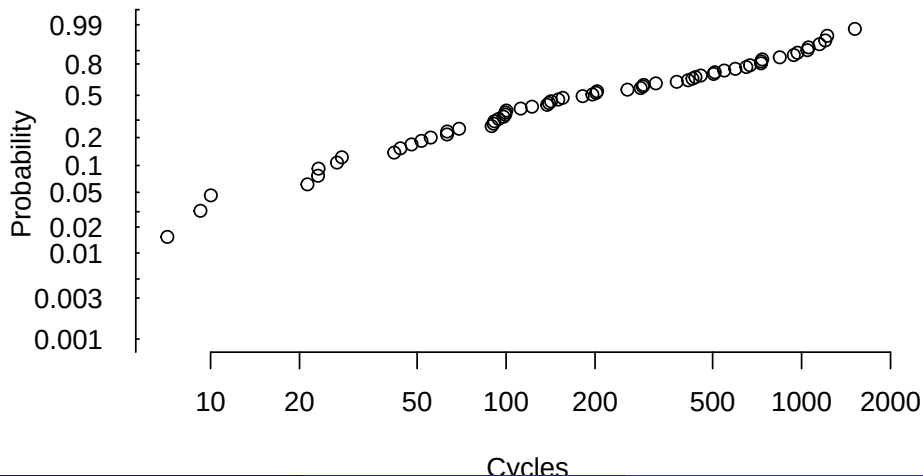
- Bottom part of the table for the Kaplan-Meier estimates

##	Cycles	F.hat.KM	S.hat.KM
## [59,]	1055.45	0.9188034	0.08119658
## [60,]	1149.66	0.9350427	0.06495726
## [61,]	1202.70	0.9512821	0.04871795
## [62,]	1221.00	0.9675214	0.03247863
## [63,]	1514.70	0.9837607	0.01623932
## [64,]	1740.75	1.0000000	0.00000000

4. Consider both Failure Mode

- Weibull Probability Plot Scale for proportion failing

Weibull Probability Scale



5. Comparing the two failure mode distributions

- Previously we assumed that each of the two failure modes followed two different distributions, Weibull and Frechet, respectively.
- Let's suppose that also the second failure mode, obstruction, follows a Weibull Distribution.
- We can perform a Likelihood ratio test, to check if under this new assumption, there is any statistically significant difference between the two failure modes

5. Comparing the two failure mode distributions

- Parameter estimates and log-likelihood for failure mode Break

```
## Call:
## survreg(formula = Surv(time = Cycles, time2 = Cycles, type
##      1, data = subset(Dishwasher, Dishwasher$Failure == fail
##      dist = "weibull"))
##
## Coefficients:
## (Intercept)
##      6.692854
##
## Scale= 0.5553359
##
## Loglik(model)= -184.1    Loglik(intercept only)= -184.1
## n= 25
```

5. Comparing the two failure mode distributions

- Parameter estimates and log-likelihood for failure mode Obstruction

```
## Call:
## survreg(formula = Surv(time = Cycles, time2 = Cycles, type
##      1, data = subset(Dishwasher, Dishwasher$Failure == fail
##      dist = "weibull")
##
## Coefficients:
## (Intercept)
##      5.064521
##
## Scale= 1.076193
##
## Loglik(model)= -243.9    Loglik(intercept only)= -243.9
## n= 40
```

5. Comparing the two failure mode distributions

- For the marginal distribution of failure mode break, we have that the maximum likelihood is -184.1,
- For the marginal distribution of failure mode obstruction the maximum likelihood is -243.9.
- The sum of the two equals -428.
- Next we fit distributions with the same sigma parameter for both.

5. Comparing the two failure mode distributions

```
## Call:
## survreg(formula = surv_object3 ~ as.factor(Failure), data = 
##      dist = "weibull")
##
## Coefficients:
##              (Intercept) as.factor(Failure)Obstruction
##              6.590515              -1.436178
##
## Scale= 0.9138773
##
## Loglik(model)= -433.1   Loglik(intercept only)= -449.7
##  Chisq= 33.22 on 1 degrees of freedom, p= 8.23e-09
## n= 65
```

5. Comparing the two failure mode distributions

- Equal sigma distributions gives a maximum likelihood equal to -433.1.
- Then the $LR_{test} = 10.2$, whereas $chisq(0.95, 1) = 3.841$.
- Since $LR_{tests} > chisq(0.95, 1)$, then we reject the hypothesis that both distributions have equal sigmas.
- Therefore, the two failure modes have statistically significant differences between each-other.

6. Conclusions

- The Mean Time To Failure of spray arms due to breaks was 790.6 cycles.
- For obstructions the MTTF did not exist under the Frechet, distribution.
- From Kaplan-Meier estimators:
 - About 95% of the spray arms survived breaks for at least 141.9 cycles
 - 95% of spray arms survived obstruction for at least 10.2 cycles.

6. Conclusions

- About 95% of spray arms survived both breaks and obstructions for at least 10.2 cycles.
- From our results it seems that the obstruction failure mode needs more attention from the dishwasher engineers.
- They should focus on finding solutions on how to make the spray arm more resistant to obstructions.

7. The End

Thank you!